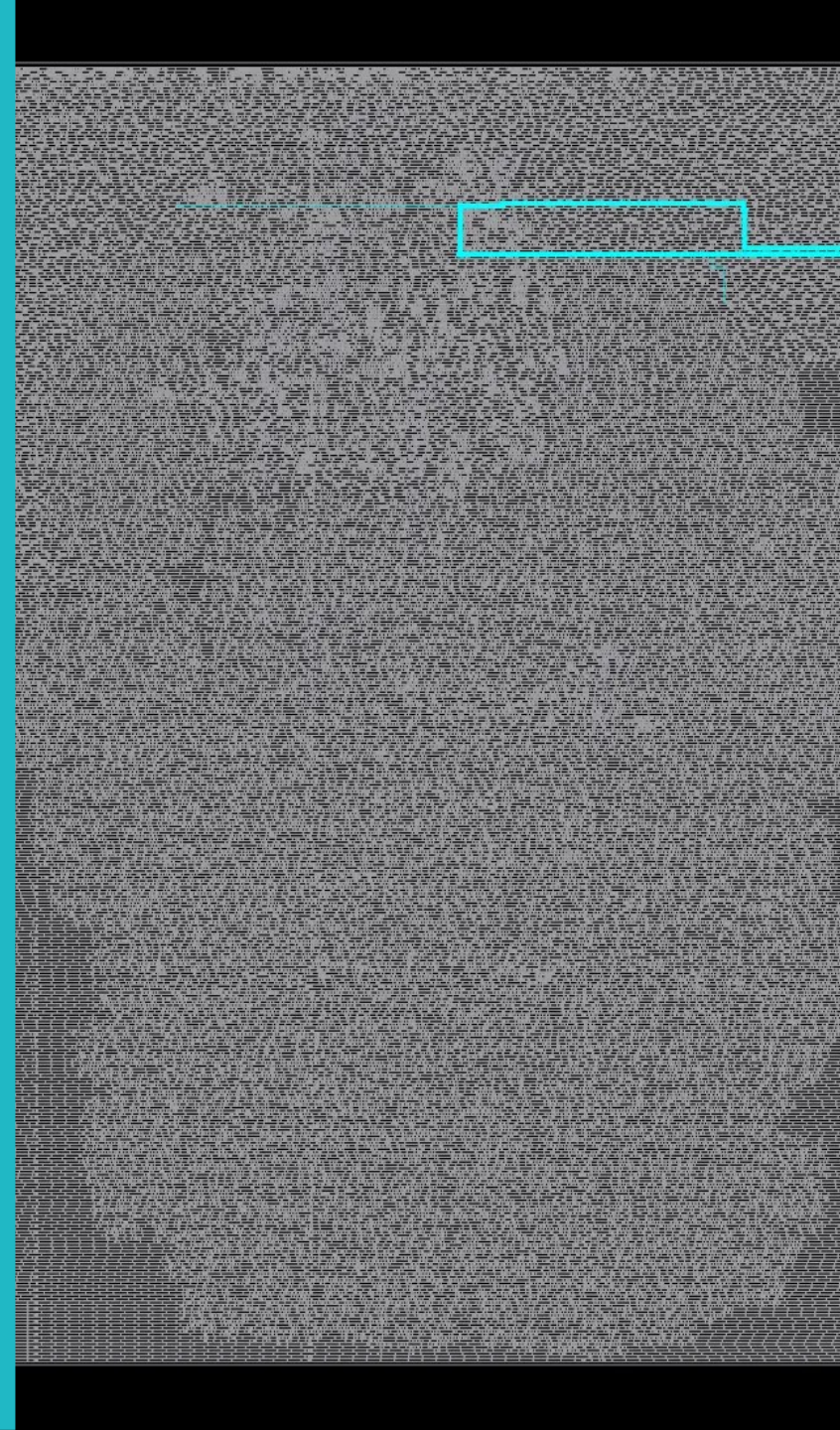


NEBULA

Constraint optimization through RTL enhancement using
Generative AI

+0.094790 ns routed WNS

Whole-design EQY PASS | 57,608-cell baseline



AI proposals need deterministic proof

A plausible RTL rewrite can silently change reset behavior, latency, CDC semantics, or physical QoR.

- **Model role** Generate a bounded, structured RTL patch around a mapped critical module.
- **Tool role** Reject out-of-scope edits, synthesis failures, timing regressions, and unproven behavior.
- **Acceptance** Require cycle-exact whole-design proof and a matched post-route improvement.

The model proposes. Measurements decide.

Five asynchronous domains at the requested scale

Requirement	Measured evidence
Master clocks	WB, TX, RX, AUX0, AUX1
Generated clocks	Five divide-by-two generated clocks
CDC	Protected two-flop toggle synchronizers
Clock division	Fixed divide-by-two plus programmable MDC
Design scale	51,359 synthesis cells; 57,608 routed stdcells

0

final route DRC

0

antenna violations

10

declared clock names

57,608

routed standard cells

Locally constructed wrapper around unchanged OpenCores Ethernet MAC RTL.

Verification-gated optimization pipeline

- | | | |
|----|----------------|--|
| 01 | Analyze | Yosys + OpenSTA parse timing and map critical endpoints to RTL |
| 02 | Propose | The model returns a structured patch within editable boundaries |
| 03 | Screen | Full-design synthesis and STA reject invalid or worse candidates |
| 04 | Prove | EQY checks cycle-exact equivalence across the complete hierarchy |
| 05 | Measure | Matched OpenROAD runs decide timing and PPA acceptance |

A balanced XOR tree shortens logic depth

Baseline

$$((((((s0 \wedge s1) \wedge s2) \wedge s3) \wedge s4) \wedge s5) \wedge s6) \wedge s7)$$

Candidate

$$((s0 \wedge s1) \wedge (s2 \wedge s3)) \wedge ((s4 \wedge s5) \wedge (s6 \wedge s7))$$

Only rtl/nebula_aux_domain.v changed

- **Preserved**
Ports, clocks, reset behavior, CDC logic, state, and cycle latency.
- **Retained evidence**
One physical QoR regression was rejected; the accepted result reproduced independently.

Whole-design cycle-exact proof passed

EQY compared the complete nebula_ethmac5_top hierarchy under the same reset sequence.

Successfully proved designs equivalent

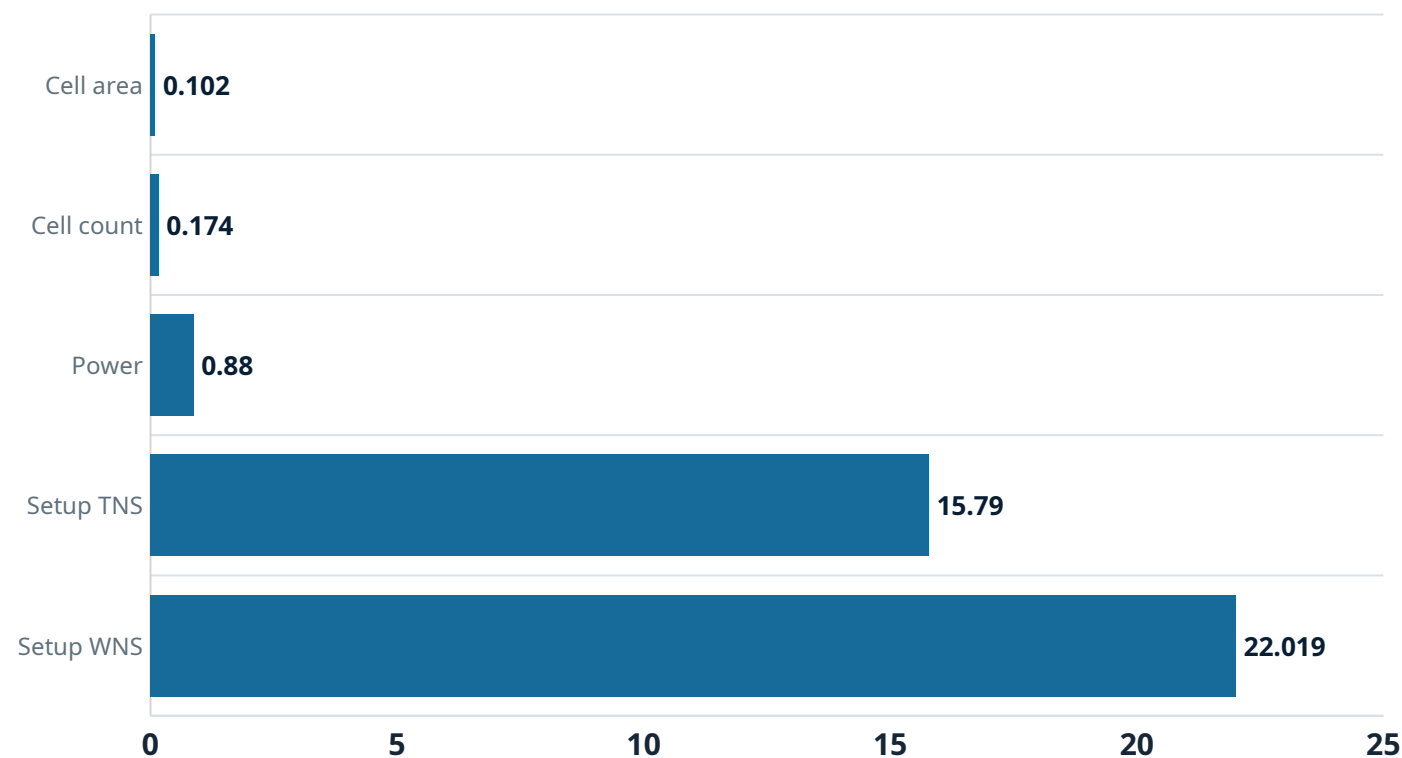
DONE (PASS, rc=0)

Elapsed clock time: 0:02:20 (140 seconds)

The state machine blocks UNKNOWN and FAIL results from acceptance.

Full 4,780-line proof log included in 04_PROOFS/formal/

Timing improved while area and power decreased



+0.094790 ns

setup WNS

+3.4968 ns

setup TNS

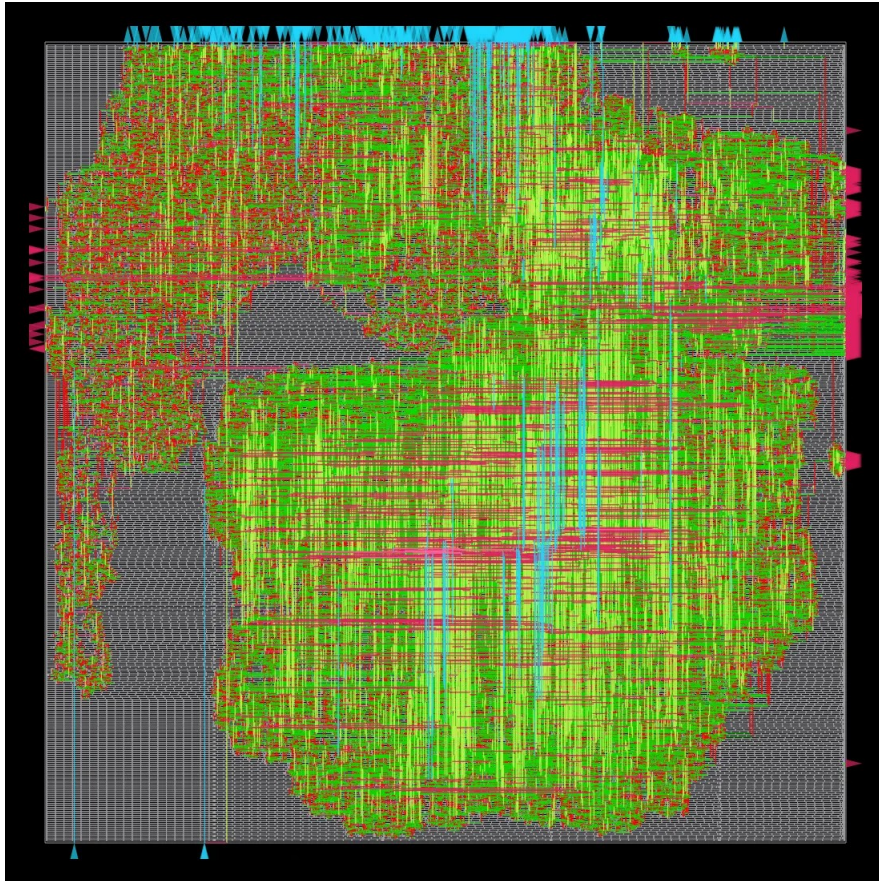
-100 cells

-142 μm^2 area

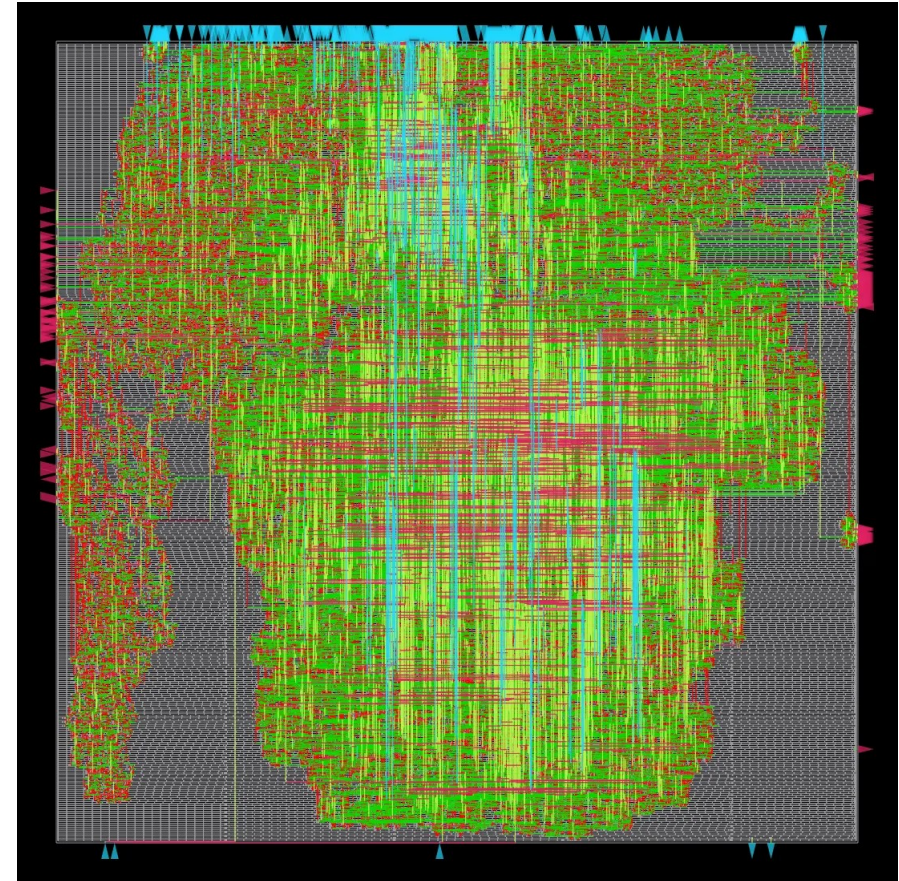
-0.88% power

matched vectorless estimate

Both designs completed detailed routing cleanly



BASELINE | 57,608 standard cells



CANDIDATE | 57,508 standard cells

Final route DRC: 0 | Antenna violations: 0

Verified optimization with explicit signoff limits

What is proved

- **Equivalence** Complete design, cycle-exact EQY PASS.
- **Physical QoR** Matched routed WNS, TNS, area, cell count, and power improved.
- **Clean routing** Zero final route DRC and zero antenna violations.

What remains

Setup WNS remains -0.335711 ns. Hold WNS remains -0.00689157 ns.

Power uses default vectorless activity. The frequency figure is an estimate, not a sweep.

Timing closure, activity-based power, and a frequency sweep remain future work.

Submission package: report + deck + RTL + exact patch + raw proofs + source archive + checksums